Exploring and Analyzing Diabetes Data Through Statistical Techniques in R

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Study Objectives

Objectives

- Explore the demographic and clinical characteristics of the diabetes dataset.
- Examine the relationship between diabetes and key risk factors.
- Evaluate statistical associations using hypothesis testing.
- Develop regression models to predict glucose levels and diabetes outcome.

Statistical Methods

- Descriptive Statistics
- Data Visualisation
- Hypothesis Testing
- Correlation Analysis
- Linear Regression
- Logistic Regression

Aim: To identify the main predictors of diabetes and evaluate their statistical significance.

"Statistics is the art and science of learning from data."
(RSS International, 2024)

Dataset Description

Dataset: Pima Indians Diabetes Database

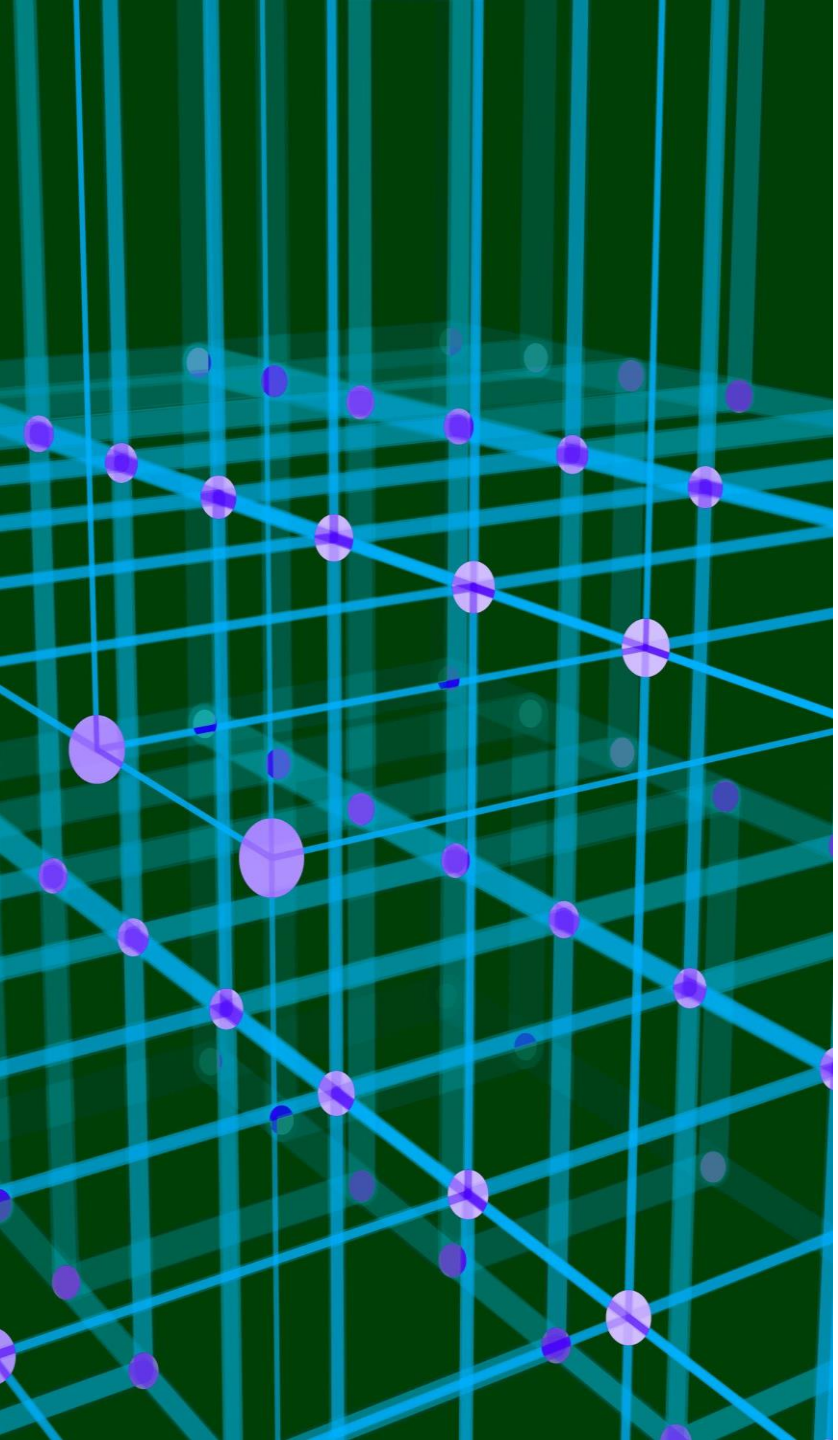
Source: National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK)

https://www.kaggle.com/datasets/akshaydattatraykhare/diabetes-dataset/data

- 768 observations
- 8 predictor variables
- 1 binary response variable (Outcome)

Variable	Description
Pregnancies	N of pregnancies
Glucose	Plasma glucose concentration
Blood Pressure	Diastolic blood pressure
Skin Thickness	Triceps skinfold thickness
Insulin	Serum insulin
BMI	Body Mass Index
DiabetesPedigreeFunction	Diabetes family history score
Age	Age in years
Outcome	0 = no diabetes 1 = diabetes

```
dim(diabetes)
names(diabetes)
str(diabetes)
```



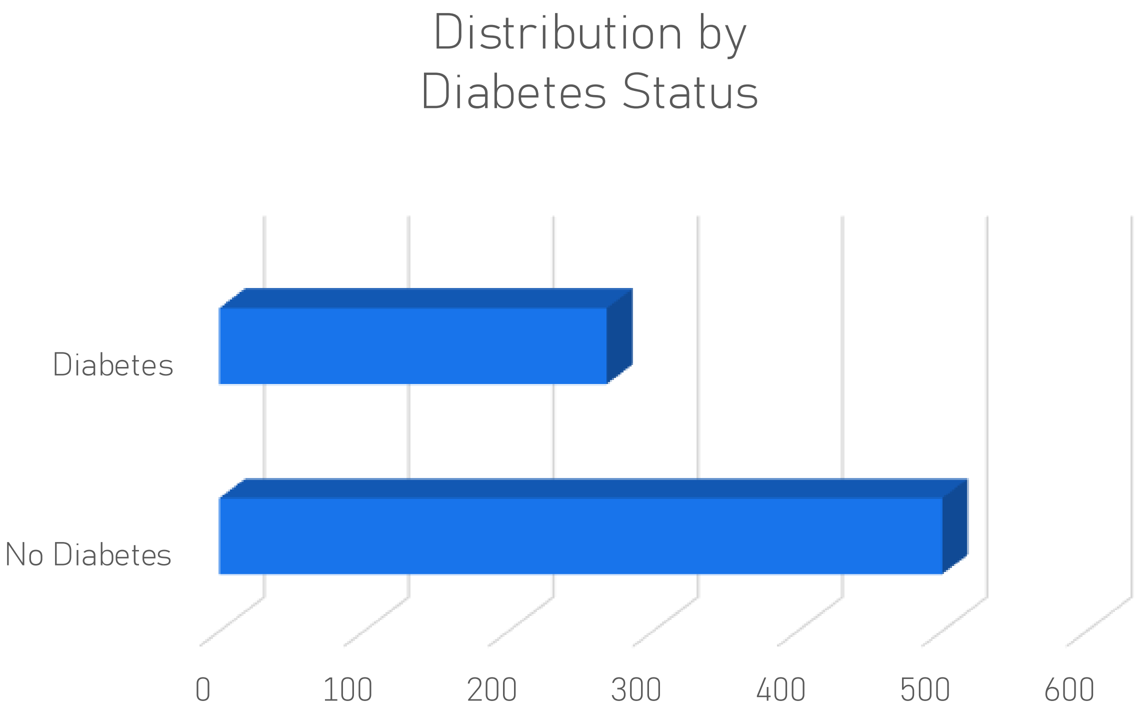
"…make both calculations and graphs. Both sorts of output should be studied; each will contribute to understanding."
Anscombe (1973)

Methodology

- Import Dataset
- Data exploration
- Descriptive Statistics
- Exploratory Visualisation
- Hypothesis Testing
- Regression Models
- Model Evaluation

library(ggplot2)
library(dplyr)

Sample Charactheristics



Characteristic	Value
Participants	768
Variables	9
No Diabetes	500 (65.1%)
Diabetes	268 (34.9%)

```
ggplot(diabetes,
     aes(x = factor(Outcome))) +
  geom_bar(fill = "#2E86C1") +
  labs(title = "Distribution by Diabetes Status")
```

Summary Statistics

•Glucose showed the greatest variability.

•BMI and Age were widely distributed.

•Summary statistics guided subsequent analyses.

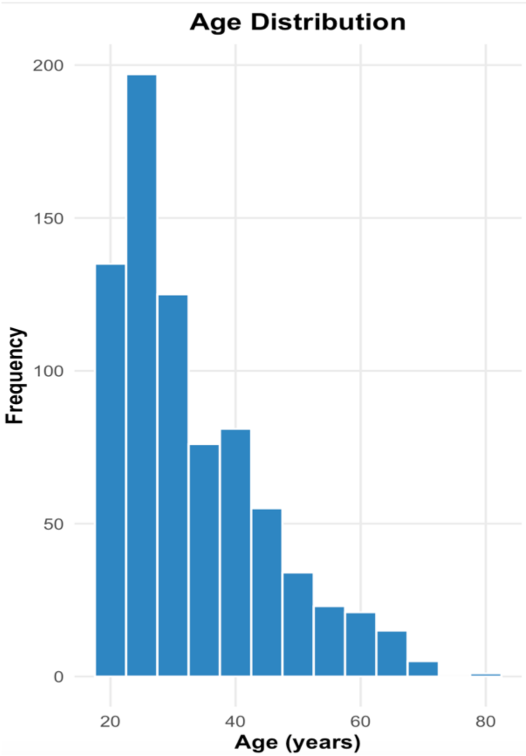
Table 1. Descriptive statistics of the main clinical variables.

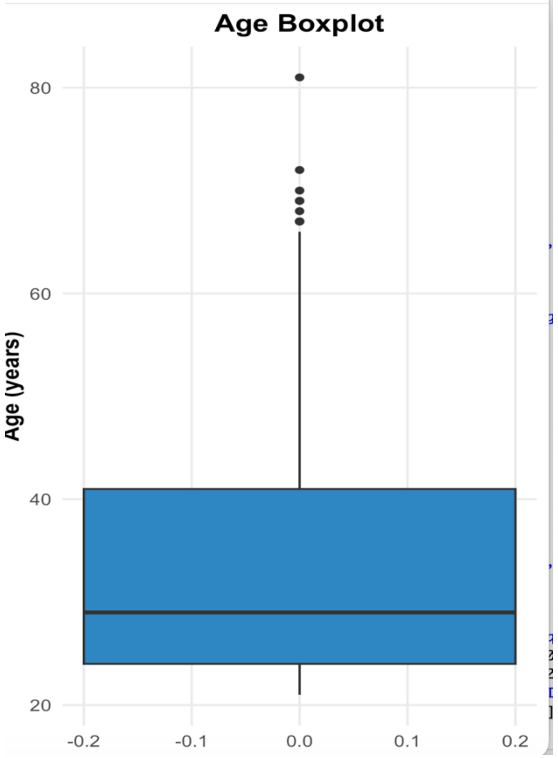
Variable	Mean	Median	Mode	SD	Min	Max
Glucose	120.89	117	100	31.97	0	199
BMI	31.99	32.0	32	7.88	0.0	67.1
Age	33.24	29	22	11.76	21	81
Blood Pressure	69.11	72	70	19.36	0	122
Pregnancies	3.85	3	1	3.37	0	17

From the Royal College of Radiologists guide:
«Descriptive statistics summarise the data before more advanced analyses. »

summary(diabetes)
describe(diabetes)

Age Distribution





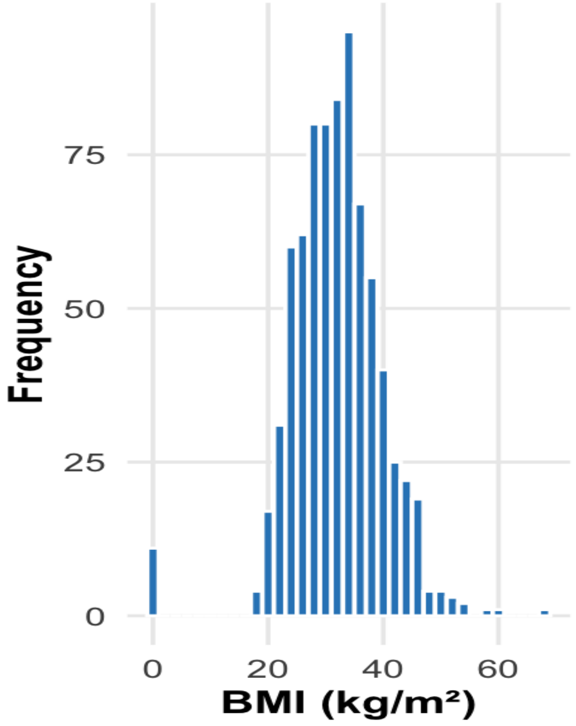
- Most participants were between 21 and 41 years.
- Median age was 29 years.
- Age distribution was positively skewed.
- Several older participants (>65 years) were identified as outliers.

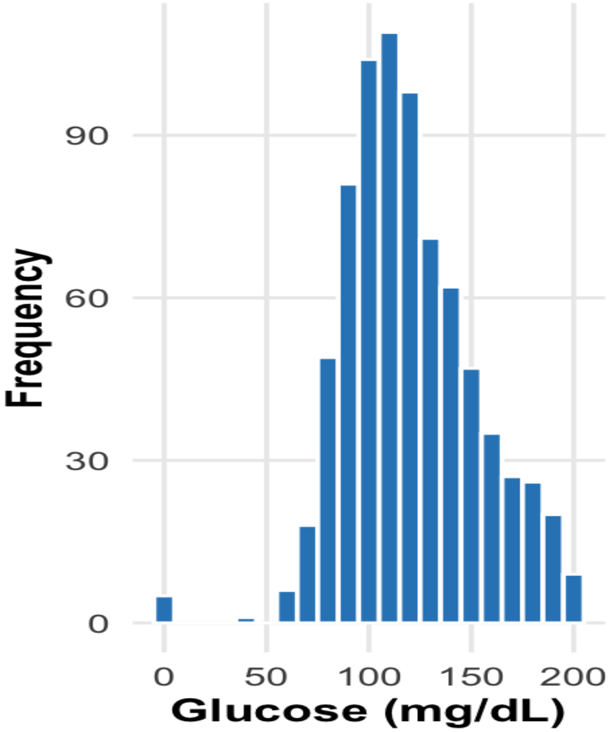
Q1	24
Median	29
Q3	41
Max	81
IQR	17

```
ggplot(diabetes, aes(x = Age)) + geom_histogram()
ggplot(diabetes, aes(y = Age)) + geom_boxplot()
```

Autodesk article:
«Similar summary statistics can hide different underlying distributions, highlighting the importance of visualising the data.»

BMI and Glucose Distribution

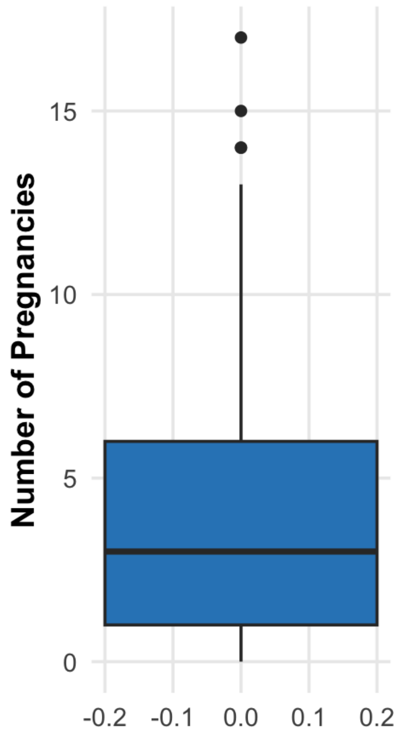


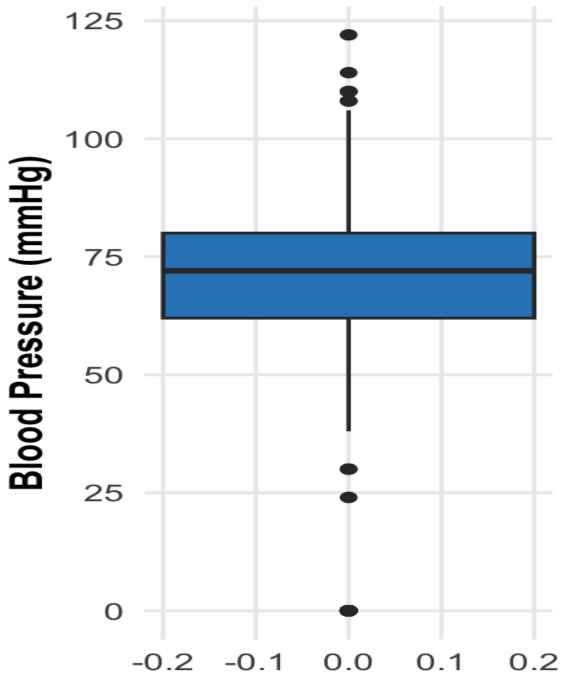


- BMI is concentrated around 30–35 kg/m².
- Glucose shows greater variability than BMI.
- Both variables exhibit right-skewed distributions.

ggplot(diabetes, aes(x = BMI)) + geom_histogram()
ggplot(diabetes, aes(x = Glucose)) + geom_histogram()

Blood pressure and pregnancies





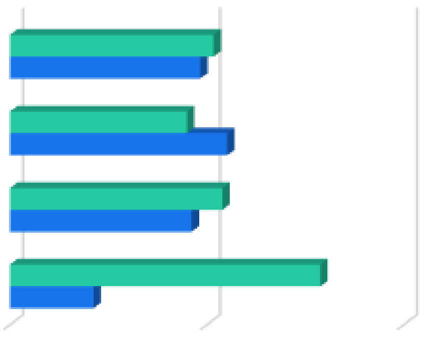
- Pregnancies shows a positively skewed distribution with several high-value outliers (14–17 pregnancies).
- Blood Pressure contains impossible values (0 mmHg), suggesting missing values were encoded as zero.
- These observations indicate that data cleaning should be considered before predictive modelling.
- Boxplots provide a simple way to detect unusual observations that may influence statistical analyses.

```
ggplot(diabetes, aes(y = Pregnancies)) +
  geom_boxplot(fill = "steelblue") +
  labs(title = "Pregnancies Boxplot")

ggplot(diabetes, aes(y = BloodPressure))
+
  geom_boxplot(fill = "steelblue") +
  labs(title = "Blood Pressure Boxplot")
```

Diabetes Rates Across Age and BMI Categories

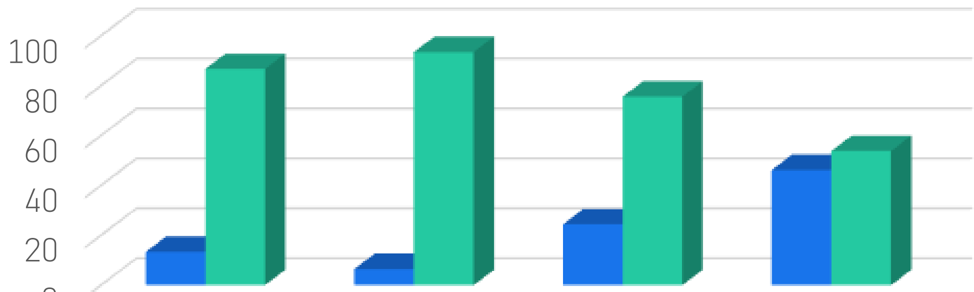
Diabetes Rate by Age Group



	<30	30-39	40-49	50+
No Diabetes	78.8	53.9	44.9	51.7
Diabetes	21.2	46.1	55.1	48.3

No Diabetes Diabetes

Diabetes Rate by BMI Category



	Underweigh t	Normal	Overweight	Obese
Diabetes	13.3	6.5	24.4	46.2
No Diabetes	86.7	93.5	75.6	53.8

Diabetes No Diabetes

```r
# Create age groups
diabetes$AgeGroup <- cut(
  diabetes$Age,
  breaks = c(20, 29, 39, 49, Inf),
  labels = c("<30", "30-39", "40-49", "50+")
)

# Create BMI categories
diabetes$BMICategory <- cut(
  diabetes$BMI,
  breaks = c(-Inf, 18.5, 25, 30, Inf),
  labels = c("Underweight", "Normal",
             "Overweight", "Obese")
)

# Calculate diabetes rates
prop.table(table(diabetes$AgeGroup,
                 diabetes$Outcome), 1)

prop.table(table(diabetes$BMICategory,
                 diabetes$Outcome), 1)
```

- Diabetes prevalence generally increased with age and BMI, with the highest prevalence observed among participants aged 40–49 and those classified as obese.

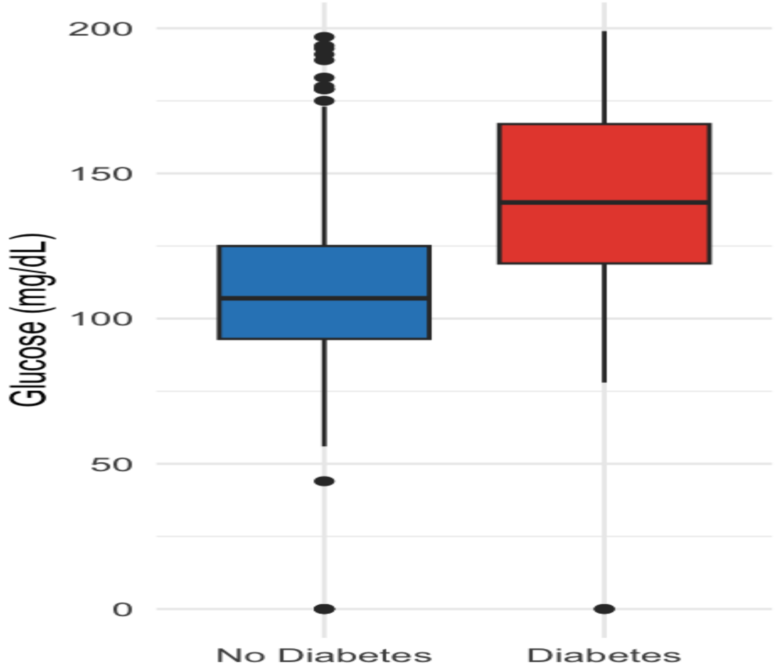
Glucose Levels by Diabetes Outcome

Key Findings

• Participants with diabetes had substantially higher glucose levels.

• Mean glucose increased from **110** to **141 mg/dL**.

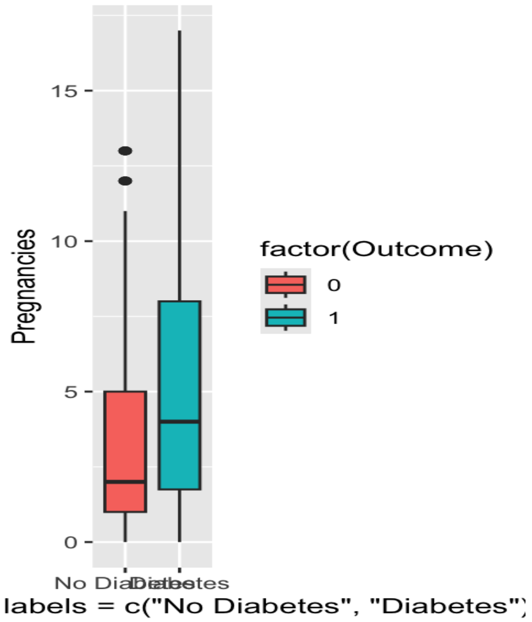
• The difference was statistically significant (**p < 0.001**).

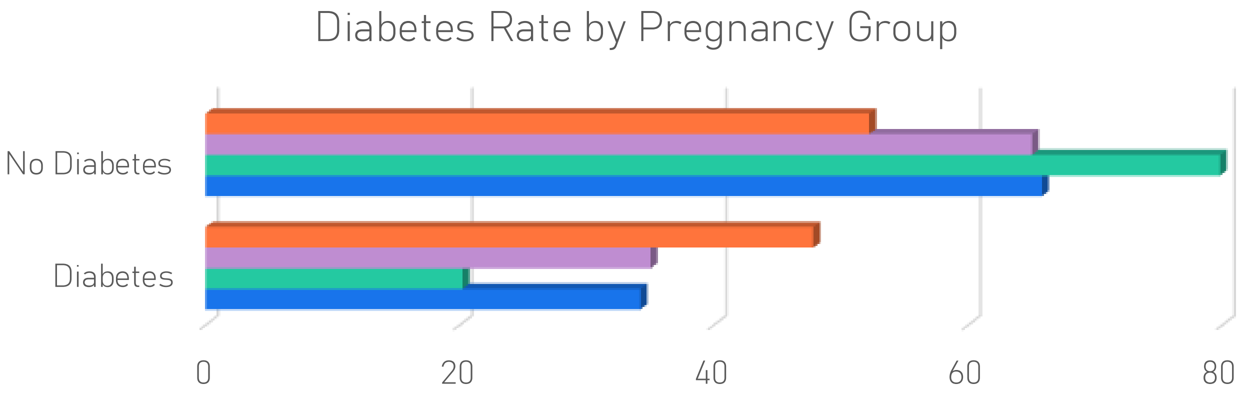



```
ggplot(diabetes, aes(x = factor(Outcome, labels =
c("No Diabetes","Diabetes")), y = Glucose, fill =
factor(Outcome))) + geom_boxplot() + labs(title =
"Glucose Levels by Diabetes Outcome", x = "", y =
"Glucose (mg/dL)") + theme_minimal() +
theme(legend.position = "none")
```

t.test(Glucose ~ Outcome, data = diabetes)

Pregnancies by Diabetes Outcome





	Diabetes	No Diabetes
5+	47.8	52.2
3-4.	35	65
1-2.	20.2	79.8
0	34.2	65.8

5+ 3-4. 1-2. 0

```
# Pregnancy groups
diabetes$PregnancyGroup <- cut(
  diabetes$Pregnancies,
  breaks = c(-1, 0, 2, 4, Inf),
  labels = c("0", "1-2", "3-4", "5+")
)

# Diabetes rates
prop.table(
  table(diabetes$PregnancyGroup,
        diabetes$Outcome), 1
)

# Group comparison
t.test(Pregnancies ~ Outcome,
       data = diabetes)
```

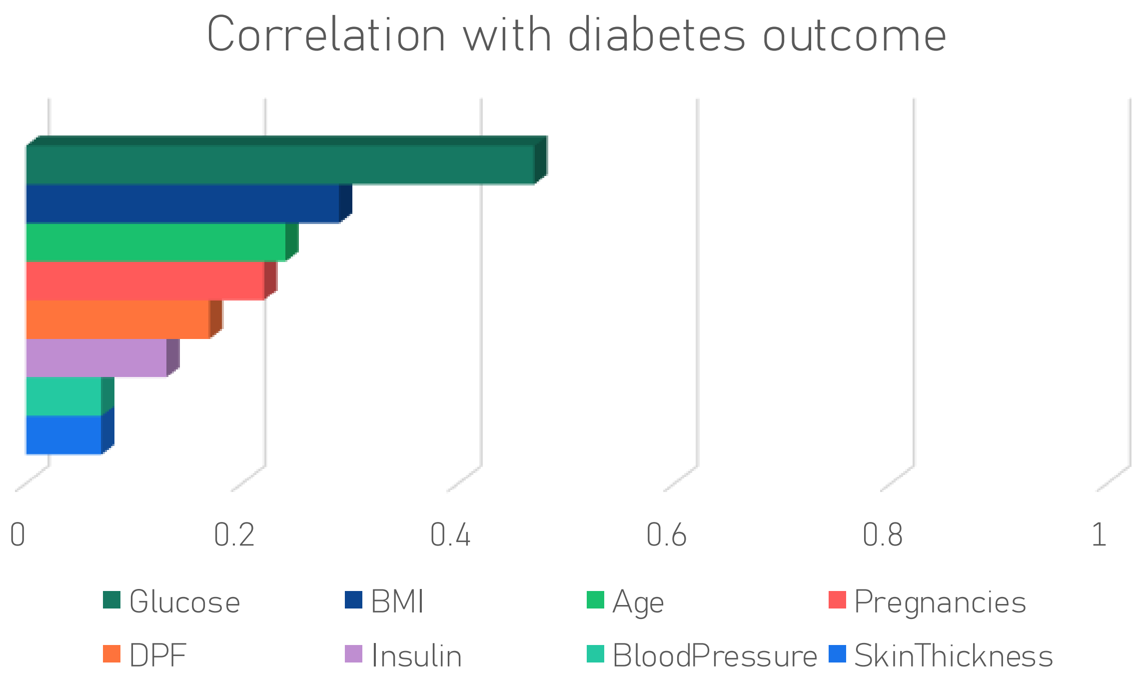
Key Findings

•Mean pregnancies: **4.87 vs 3.30**

•**Significant difference ($p < 0.001$)**

•Highest diabetes rate in **5+ pregnancies (47.8%)**

Correlation Matrix



Key Findings

•Glucose shows the strongest positive association with Outcome.

•BMI and Age exhibit moderate positive relationships.

• No strong correlations (>0.70) were observed, indicating low multicollinearity among predictors.

The correlation analysis provides the basis for the regression models presented next.

```
cor_matrix <- cor(diabetes)

round(cor_matrix, 2)
```

Hypothesis Testing

Analysis	Test	Statistic	P-value	Decision
Diabetesx age groups	Chi-square	$\chi^2 = 69.92$	<0.001	Reject H_0
DiabetesxBMI Categories	Chi-square	$\chi^2 = 76.43$	<0.001	Reject H_0
Glucose by Age Groups	ANOVA	$F = 19.62$	<0.001	Reject H_0

Diabetes prevalence differed significantly across age and BMI categories. In addition, older participants (50+) had significantly higher mean glucose levels than younger participants.

Post-hoc (Tukey HSD)

- 50+ years had significantly higher glucose levels ($p < 0.01$).
- 30-39 and 40-49 years showed no significant difference ($p = 0.998$).

```
chisq.test(table(diabetes$AgeGroup, diabetes$Outcome))

chisq.test(table(diabetes$BMICategory, diabetes$Outcome))

anova_model <- aov(Glucose ~ AgeGroup, data = diabetes)

summary(anova_model)

TukeyHSD(anova_model)
```

Multiple Linear Regression

Significant Predictors

Predictor	β Estimate	p-value	Significant
Age	+0.645	<0.001	✓
BMI	+0.750	<0.001	✓
Insulin	+0.100	<0.001	✓
Skin Thickness	-0.334	<0.001	✓
DPF	+6.316	0.046	✓
Pregnancies	+0.059	0.871	✗
Blood Pressure	+0.070	0.220	✗

Model Performance

Metric	Result
Observations	768
R^2	0.230
Adjusted R^2	0.223
F-statistic	32.46
Model p-value	< 0.001

- Model significant ($p < 0.001$)
- $R^2 = 23\%$
- Age, BMI, Insulin and DPF were significant predictors

```r
linear_model <- lm(
  Glucose ~ Age + BMI + Pregnancies +
    BloodPressure + SkinThickness +
    Insulin + DiabetesPedigreeFunction,
  data = diabetes
)

summary(linear_model)
```

Logistic Regression

Model Results

Predictor	Odds Ratio	P-value	interpretation
BMI	1.09	<0.001	Higher BMI increases diabetes risk
Age	1.03	<0.001	Older age increases diabetes risk
Glucose	1.03	<0.001	Higher glucose increases diabetes risk

Model Performance

Metric	Result
Hosmer-Lemeshow t	P = 0.349
Classification Accuracy	76.8%
Sensitivity	56.3%
Specificity	87.8%

- All three predictors (**BMI, Age and Glucose**) were statistically significant.
- The Hosmer–Lemeshow test was **not significant (p = 0.349)**, indicating that the model fits the data adequately.
- The model correctly classified **76.8%** of participants.
- The model identified **87.8% of non-diabetic individuals** (high specificity) but only **56.3% of diabetic individuals**(moderate sensitivity).

```
log_model <- glm(
  Outcome ~ BMI + Age + Glucose,
  data = diabetes,
  family = binomial
)

summary(log_model)
```

```
hoslem.test(diabetes$Outcome,
  fitted(log_model))

predicted <- ifelse(
  predict(log_model, type = "response") > 0.5,
  1, 0
)
```

Model Comparison

Logistic Models

Model	Predictors
Model 1	BMI+Age+Glucose+Pregnancies
Model 2	BMI+Age+Glucose+Pregnancies+BMIx Age

Likelihood Ratio Test

Test	Result
χ^2	0.039
P-value	0.843
Decision	Interaction not significant

Odds Ratios (Model 2)

Predictor	OR	P-value
BMI	1.08	0.064
Age	1.01	0.871
Glucose	1.03	<0.001
Pregnancies	1.12	<0.001
BMI x Age	1.00	0.843

```
model1 <- glm(
  Outcome ~ BMI + Age + Glucose + Pregnancies,
  family = binomial,
  data = diabetes
)
model2 <- glm(
  Outcome ~ BMI * Age + Glucose + Pregnancies,
  family = binomial,
  data = diabetes
)
anova(model1, model2, test = "Chisq")
```

- Adding the **BMI × Age interaction** did **not** improve the model (Likelihood Ratio Test: **p = 0.843**).
- The interaction term was **not statistically significant**.
- **Glucose** and **Pregnancies** remained significant predictors.
- **Model 1** was preferred because it was **more parsimonious** while providing a similar fit.

Discussion

Key Findings
Diabetes prevalence increased with **age** and **BMI**.
Individuals with diabetes had significantly **higher glucose levels**.
Women with diabetes reported **more pregnancies** on average.
Glucose, BMI and Age were significant predictors of diabetes.
The **BMI × Age interaction** was not statistically significant.

Comprehensive statistical analysis helps identify patterns and relationships within health datasets, improving interpretation and supporting evidence-based conclusions.

Age and diabetes

Older adults had higher glucose levels and higher diabetes prevalence, consistent with previous epidemiological studies.

BMI as a risk factor

Higher BMI significantly increased diabetes risk, supporting evidence that obesity is a major contributor to Type 2 diabetes

Glucose

Glucose was the strongest predictor in both regression models, which agrees with established clinical knowledge.

Limitations & Future Work

Future Work

✓ Data cleaning & missing value imputation

✓ Additional clinical & lifestyle variables

✓ Machine learning models (Random Forest, XGBoost)

✓ External dataset validation

Limitations

❌ Zero values treated as real observations

❌ Cross-sectional dataset

❌ Limited predictor variables

❌ Moderate model performance (R^2 = 23%)

Future studies should use larger, cleaner datasets and more advanced predictive models.

Conclusion

Key Findings
768 participants were analysed.
34.9% were diagnosed with diabetes.
Age, BMI and Glucose were the strongest factors associated with diabetes.
Significant differences were found in **glucose levels** and **number of pregnancies** between diabetic and non-diabetic individuals.
Logistic regression achieved **76.8% classification accuracy**.

Statistical modelling successfully identified key diabetes risk factors and demonstrated the value of regression techniques for healthcare data analysis.

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